



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Using API will request the data through URL
- Using Data wrangling, process the data in a way we needed
- Extract the necessary data from the dataset using SQL
- Visualized data so that audience can understand the data
- Build the interactive map
- Build the dashboard

Introduction

- The data is mixed with the other payload model such as Falcon 1
- Deliver data driven insights to determine if the first stage of Falcon 9 will land successfully

Section 1

Methodology

Methodology

Executive Summary

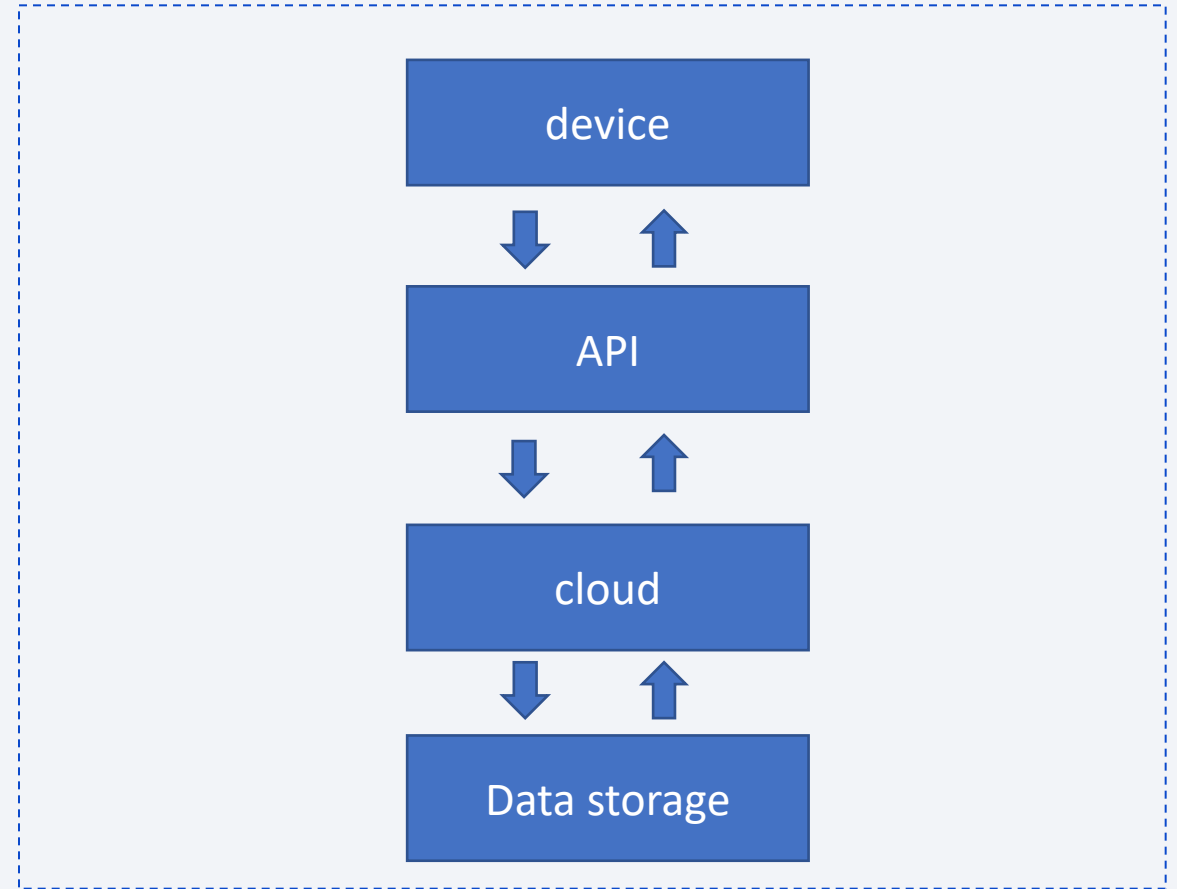
- Data collection methodology:
 - Using SpaceX Rest API
- Perform data wrangling
 - Using API, get the response and transform from json into useful data
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Used a predictive model such as SVM, logistic regression and etc to find the best model

Data Collection

- Data was collected using SpaceX rest API about launches:
 - Rocket used
 - Payload delivered
 - Launch specifications
 - Landing outcomes
 - And more

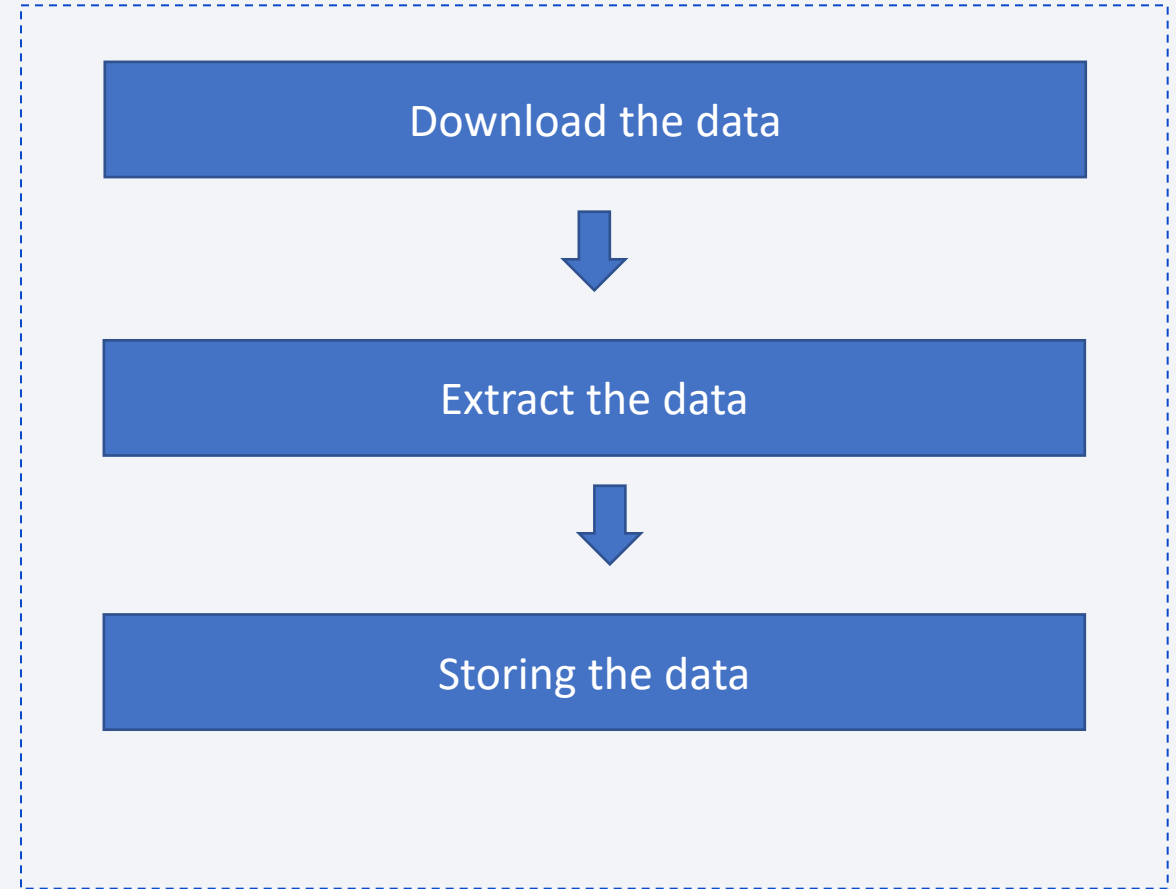
Data Collection – SpaceX API

- Present your data collection with SpaceX REST calls using key phrases and flowcharts
- Add the GitHub URL of the completed SpaceX API calls notebook (must include completed code cell and outcome cell), as an external reference and peer-review purpose



Data Collection - Scraping

- Present your web scraping process using key phrases and flowcharts
- Add the GitHub URL of the completed web scraping notebook, as an external reference and peer-review purpose



Data Wrangling

Looking at the detailed data

- Replace the null value with the mean value of the column
- Classify landing outcomes as success and failure and add the value into the new column 'Class'
- Which helps to overlook the success rate of the rocket
- Look for a detailed understanding of each column.
 - Such as Orbit what is LEO, VLEO, GTO, etc

EDA with Data Visualization

- Scatter plot payload mass vs launch site
 - For CCAFS SLC 40, it tends to have less payload mass
 - For VAFB SLC 4E, it has only 5 points but it varies from 0 to 10000
 - For KSC LC 39A, it varies from 2000 to 16000
- Scatter plot flight number vs launch site
 - For CCAFS SLC 40, from 0-30 and from 40 it is used frequently
 - For VAFB SLC 4E only used time to time from 0 to 60
 - For KSC LC 39A started to be used around 25 and used from then

EDA with SQL

- Find the unique launch sites
- Find the launch site name beginning with “CCA”
- Find the total payload mass carried by boosters from NASA
- Find the average payload mass by F9 v1.1
- Find the dates of the first successful landing outcome on a ground pad
- Find the name of the boosters that have successfully landed on a drone ship and had payload mass greater than 4000 but less than 6000

Build an Interactive Map with Folium

- Add the Circle around the launch sites
- Add the marker to the nearest cost line (In my case, I made a marker at the middle of the ocean so that I can see the clear line)
- Add the line between the marker and one of the launch sites

Build a Dashboard with Plotly Dash

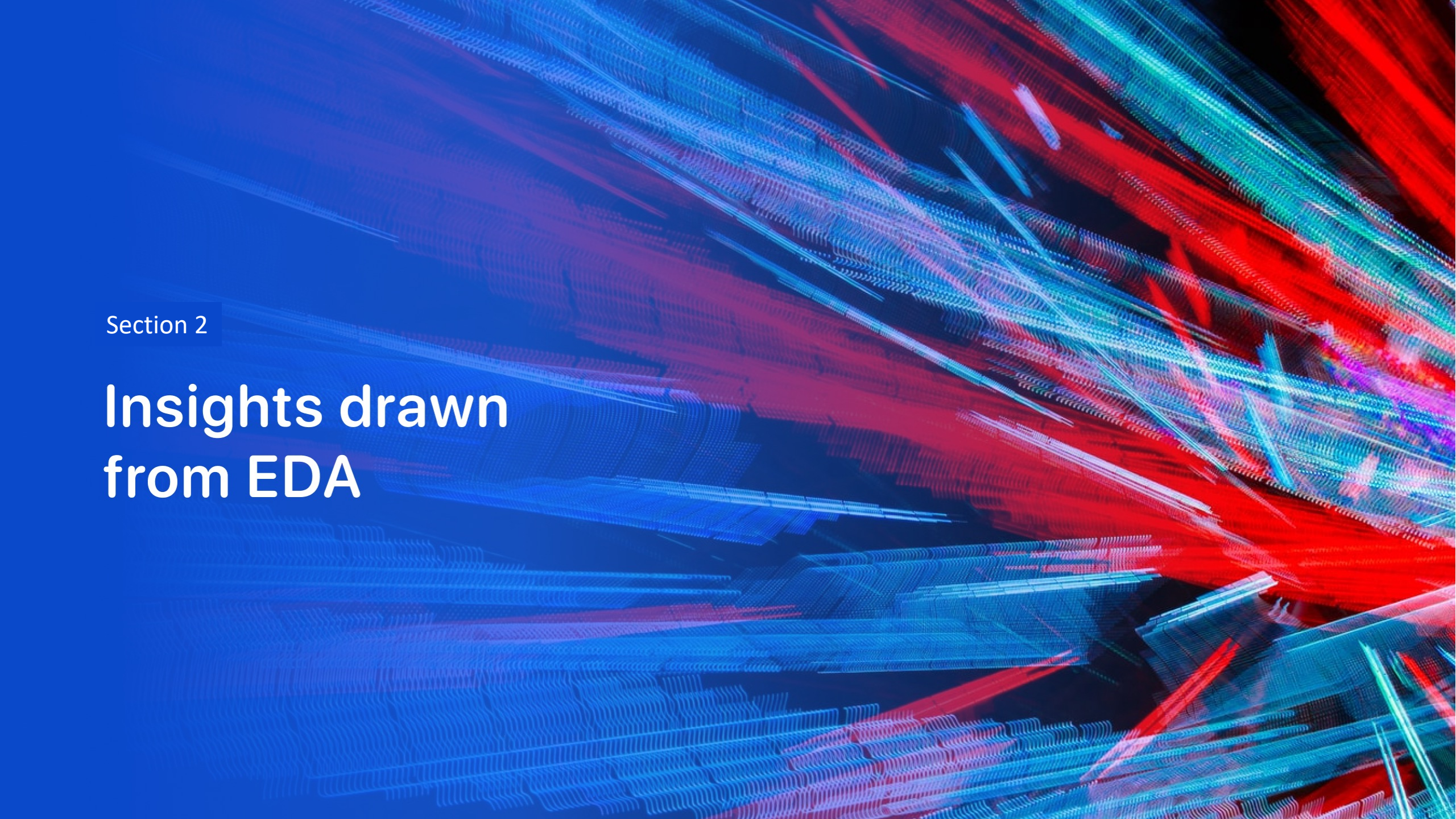
- Summarize what plots/graphs and interactions you have added to a dashboard
- Explain why you added those plots and interactions
- Add the GitHub URL of your completed Plotly Dash lab, as an external reference and peer-review purpose

Predictive Analysis (Classification)

- Calculate the accuracy for each model
- Visualize the model accuracy for
 - Logistic regression model
 - Support vector machine
 - Decision tree classifier object

Results

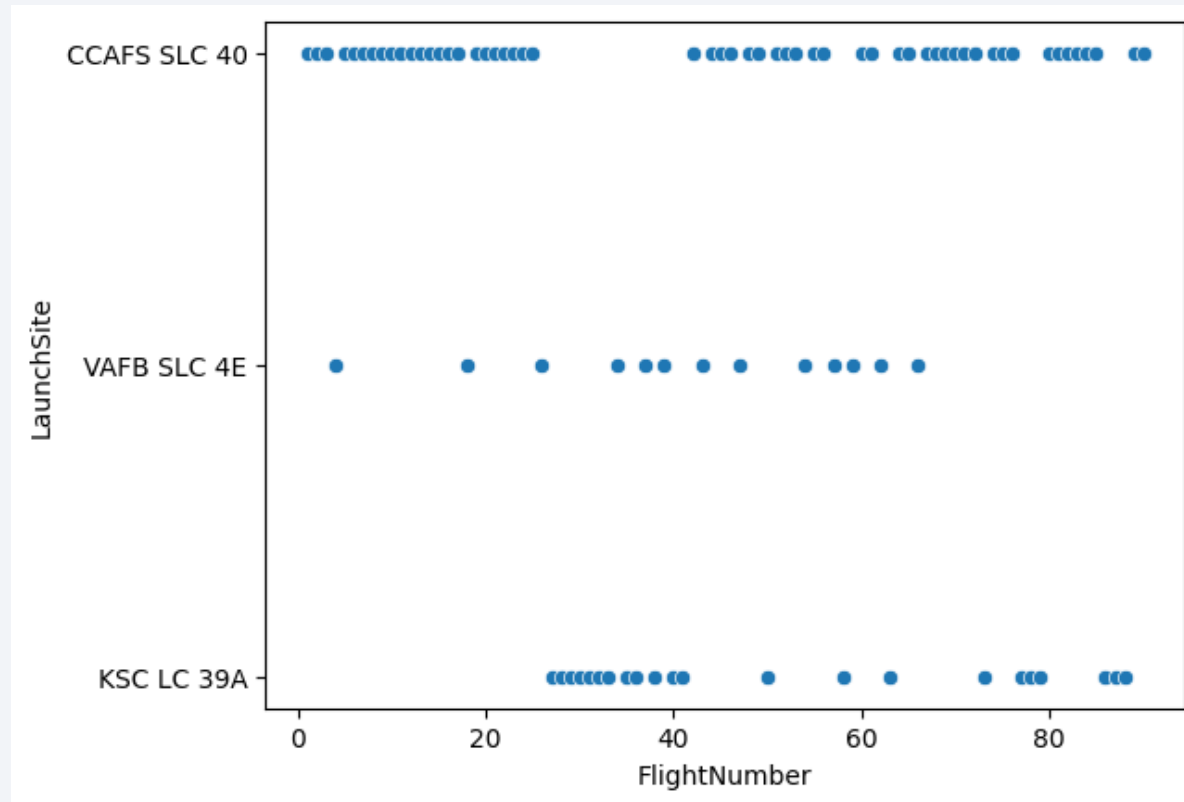
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

The background of the slide is a complex, abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks and lines in shades of red, teal, and light blue, creating a sense of motion and depth. A faint, grid-like pattern is also visible, particularly in the lower right quadrant, suggesting a digital or data-driven theme.

Section 2

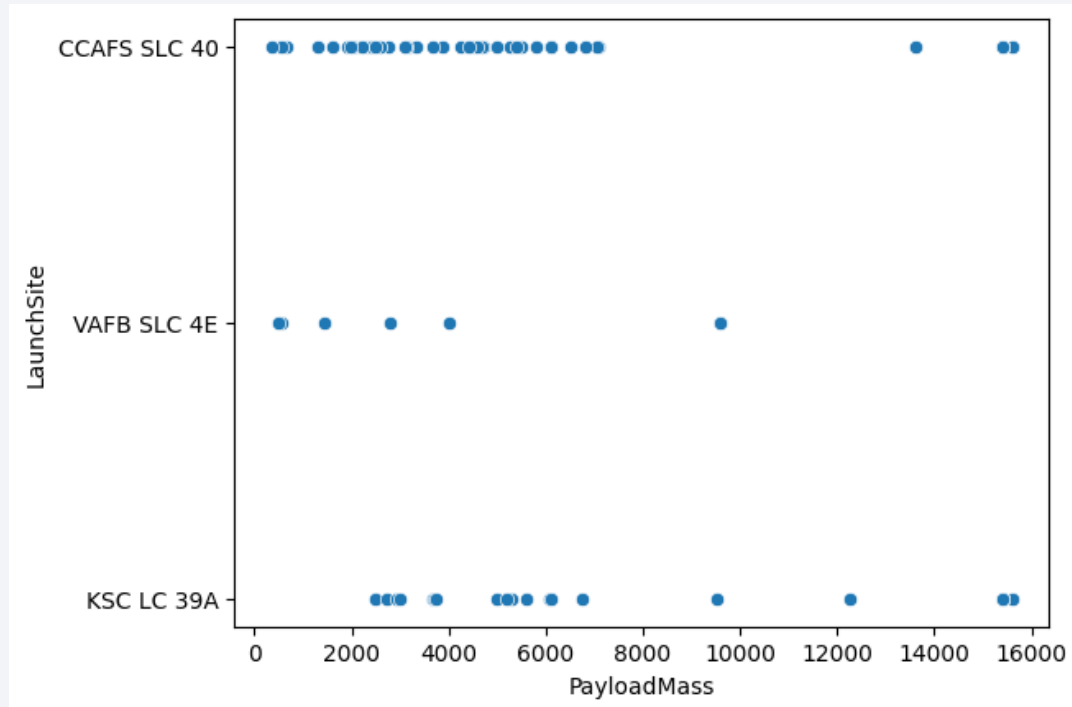
Insights drawn from EDA

Flight Number vs. Launch Site



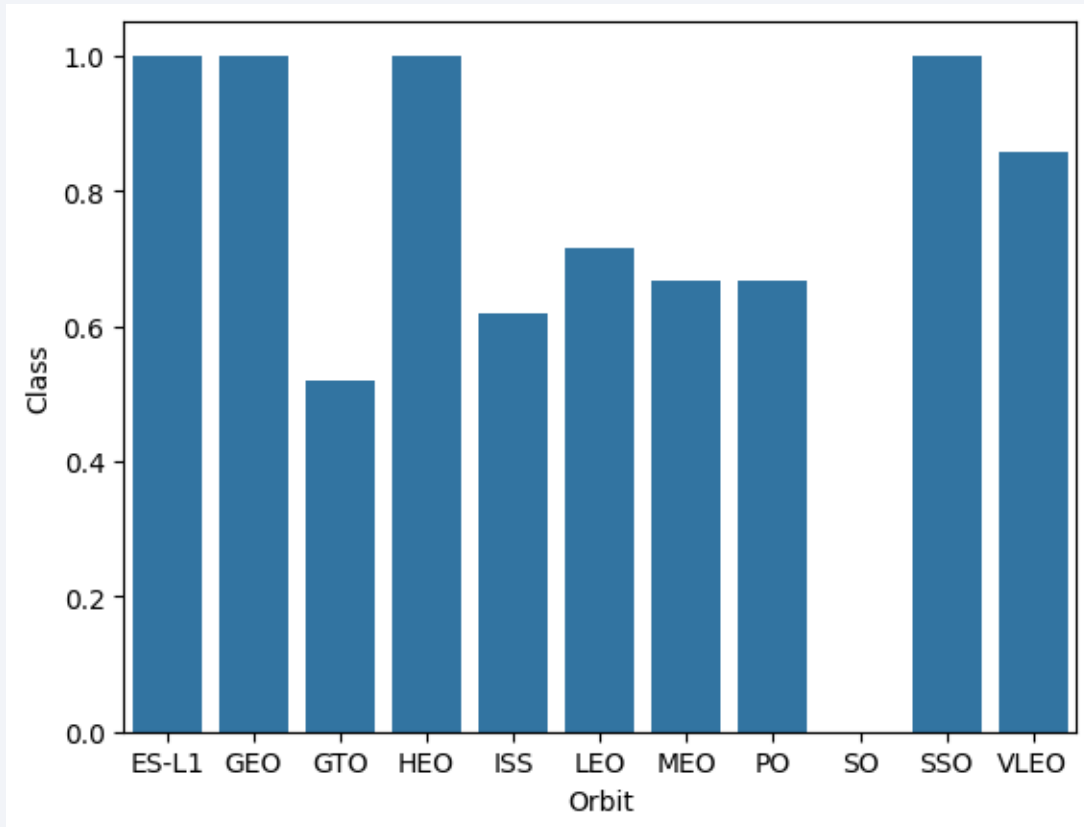
- Most of the lunch were held on first lunch site “CCAFS SLC 40”
- Second lunch site “VAFB SLC 4E” only made occasionally through out flight number
- Third lunch site “ KSC LC 39A” were started to lunch from flight number starting from around 25.

Payload vs. Launch Site

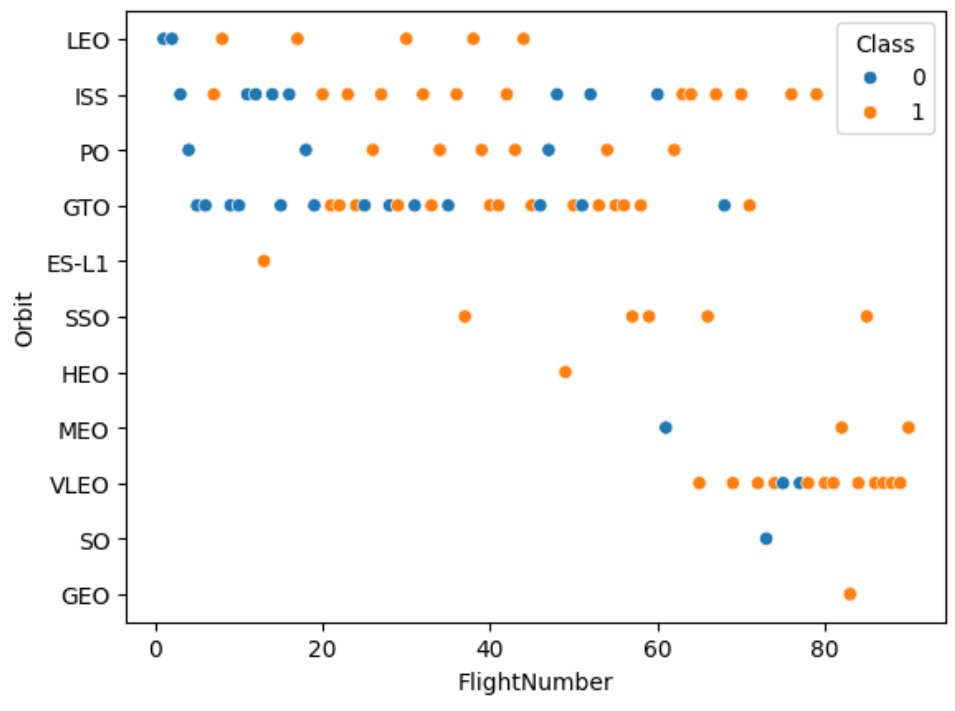


- From the first launch site “CCAFS SLC 40” most of the payload mass was below 8000kg while only 3 payload was out of bound.
- Second launch site “ VAFB SLC 4E” were tend to have distributed payload from 0 to 10000kg.

Success Rate vs. Orbit Type

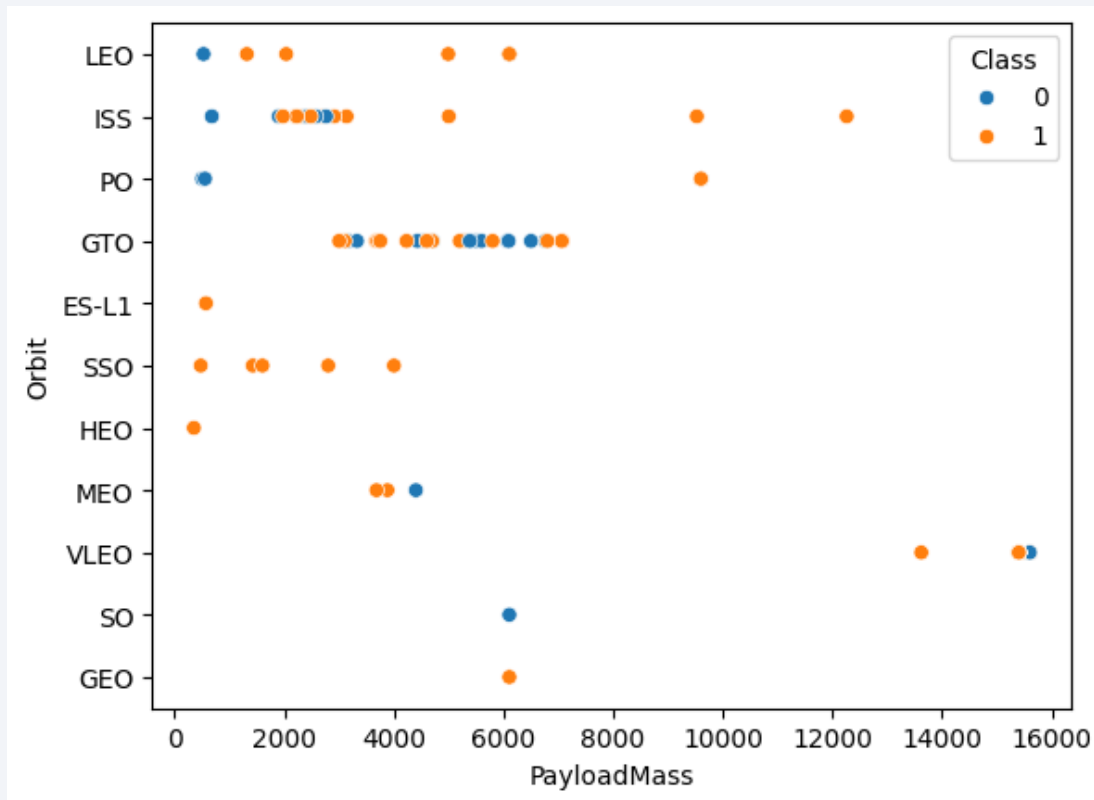


- The success rate of orbit ES-L1, GEO, HEO, SSO was 100% for all the lunches.
- Orbit GTO, ISS, LEO, MEO, PO success rates varied from 50% to 70% which is pretty low compared to the orbits above which scores 100% success rate



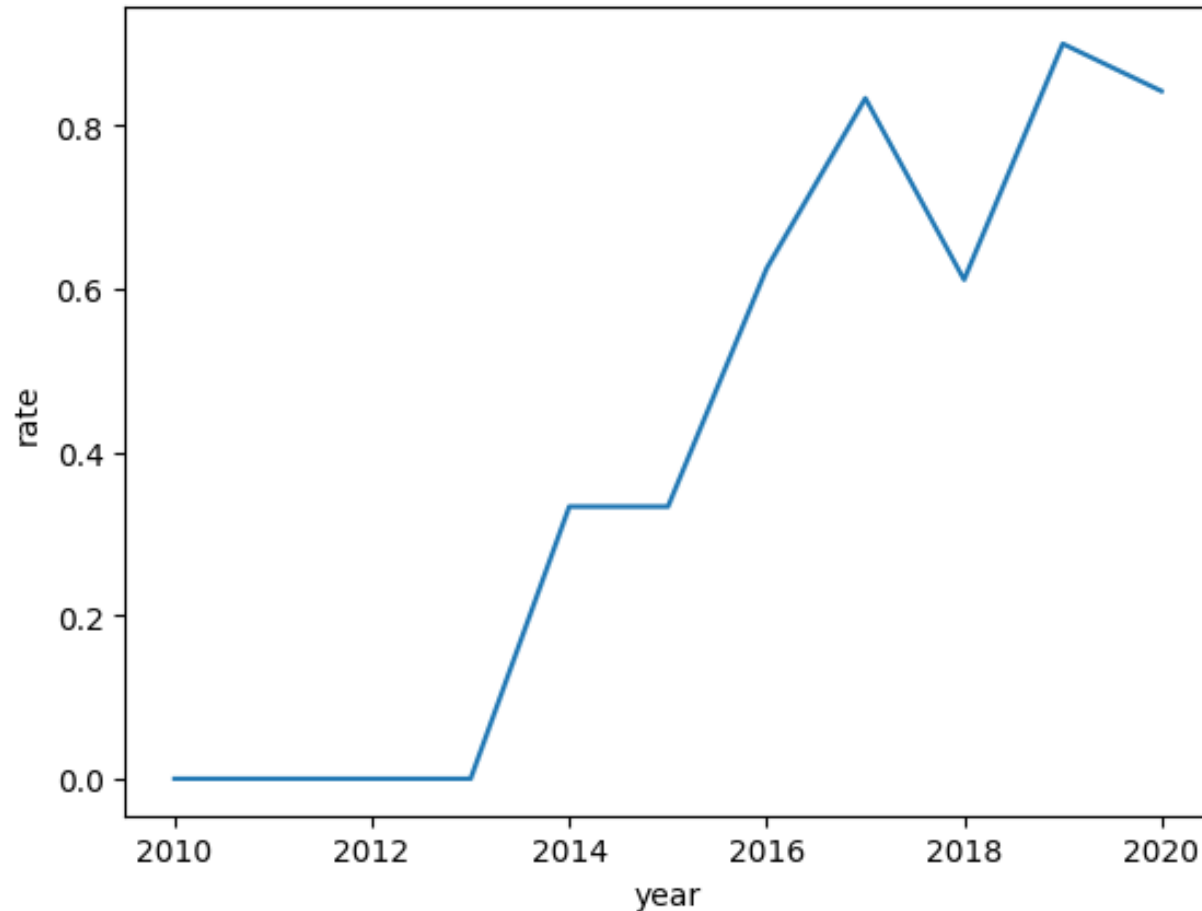
- When the flight number increases, the success rate increases as well.
- For example, from flight number 60 till the end, only around 5 failures were found.
- While flight numbers from 0-20, there are more failures than successes.

Payload vs. Orbit Type



- From the plot above orbit vs success rate, GTO was doing worst and it seems reasonable that GTO has to highest density of the payload among the other orbit
- The failures seem to vary all over the payload mass which might suggest that payload mass and success might not be correlated

Launch Success Yearly Trend



- The line plot clearly shows that the success rate increased by the year which was 0% from 2010 to 2013.
- The success rate increases sharply from 2013 to 2017 reaching 80% from 0%.
- From 2017 to 2020, the success rate varies from 60% to 90% which is a clear increase from 2010 to 2017.

All Launch Site Names

```
%sql select distinct Launch_Site from SPACEXTABLE
```

```
* sqlite:///my_data1.db  
one.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

- All launch site is “CCAFS LC -40”, “VAFB SLC-4E”, “KSC LC-39A”, “CCAFS SLC-40”
- Used “distinct” query to extract the unique value from the columns “Launch_Site” from dataset SPACEXTABLE

Launch Site Names Begin with 'CCA'

```
File display  
select * from SPACEXTABLE where Launch_Site like 'CCA%' limit 5
```

```
* sqlite:///my_data1.db  
one.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_O
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (par
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (par
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No

- Used like “CCA%” that starts with CCA and limits 5 to present only the first 5 results.

Total Payload Mass

```
%sql select sum(PAYLOAD_MASS__KG_) as total_mass from SPACEXTABLE where Customer like 'NASA%'
```

```
* sqlite:///my_data1.db  
Done.
```

total_mass

99980

- Used sum() query to sum all the elements in the column “PAYLOAD_MASS” and used “as” query to name the new column as “total_mass”
- Used “NASA%” to fine the result start with NASA

Average Payload Mass by F9 v1.1

```
%sql select avg(PAYLOAD_MASS__KG_) as average_mass from SPACEXTABLE where Booster_Version like 'F9%'
```

```
* sqlite:///my_data1.db  
Done.
```

average_mass
6138.287128712871

- Calculated the average payload mass carried by booster version F9 v1.1
- Used avg() query to average the values from the column "PAYLOAD_MASS__KG_) and assigned the new name as "average_mass" by using the "as" query.
- Also used the "like" query with "F9%" so that capture the column name "Booster_Version" start with F9.

First Successful Ground Landing Date

```
%sql select min(Date) from SPACEXTABLE where Landing_Outcome like 'Success'
```

```
* sqlite:///my_data1.db
```

```
Done.
```

min(Date)

2018-07-22

- Found the dates of the first successful landing outcome on ground pad
- Used the “min()” query to min value of the “Date” column which would be considered as the first date where the “Landing_Outcome” is like “Success”

Successful Drone Ship Landing with Payload between 4000 and 6000

```
%sql select distinct(Booster_Version) from SPACEXTABLE where Landing_Outcome
is 'Success (drone ship)' and (PAYLOAD_MASS__KG_ >4000 and PAYLOAD_MASS__KG_ <6000)

* sqlite:///my\_data1.db
Done.
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

- Listed the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000
- Used after “where”, added “is” to indicate the columns and specify the range needed, which is 4000 to 6000 using and.
- *(The screenshot is different because the query was too long to capture from GitHub so I have to capture it from visual studio)*

Total Number of Successful and Failure Mission Outcomes

```
%sql select count(*) as count from SPACEXTABLE group by Mission_Outcome is 'Success'
```

```
* sqlite:///my_data1.db  
done.
```

count

3

98

- Calculated the total number of successful and failure mission outcomes
- Used “count()” function using “*” to count every rows from SPACEXTABLE and group them by “Mission_Outcome” columns where it is success

Boosters Carried Maximum Payload

```
%sql select Booster_Version From SPACEXTABLE where max(PAYLOAD_MASS__KG_)
```

```
* sqlite:///my_data1.db
```

```
sqlite3.OperationalError) misuse of aggregate function max()
```

```
SQL: select Booster_Version From SPACEXTABLE where max(PAYLOAD_MASS__KG_) ]
```

```
Background on this error at: http://sqlalche.me/e/e3q8)
```

```
%sql select * from (select max(PAYLOAD_MASS__KG_) from SPACEXTABLE)
```

```
* sqlite:///my_data1.db
```

```
one.
```

max(PAYLOAD_MASS__KG_)

15600

- Listed the names of the booster which have carried the maximum payload mass
- First, extracted the “Booster_Version” with the maximum “PAYLOAD_MASS_KG_” and present the element

2015 Launch Records

```
%sql select substr(Date,6,2) as month_names, Landing_Outcome as failure_Landing_Outcome, Booster_Version, Launch_Site, Date
from SPACEXTABLE where Date like '2015%' and Landing_Outcome like 'Failure%'
```

Python

```
* sqlite:///my\_data1.db
```

Done.

month_names	failure_Landing_Outcome	Booster_Version	Launch_Site	Date
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40	2015-01-10
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40	2015-04-14

- Listed the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015
- First, extract the Year from “Date” as “month_names”
- Extract the “Landing_Outcome” as “failure_Landing_Outcome”, and more columns from SPACEXTABLE, used “like” function to extract 2015 and failure from “Date” and “Landing_Outcome”

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
%sql select Landing_Outcome, count(*) as count from SPACEXTABLE where (Date > '2010-06-04' and Date < '2017-03-20')  
group by Landing_Outcome order by count desc
```

✓ 0.0s

Python

* [sqlite:///my_data1.db](#)
Done.

Landing_Outcome	count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Precluded (drone ship)	1
Failure (parachute)	1

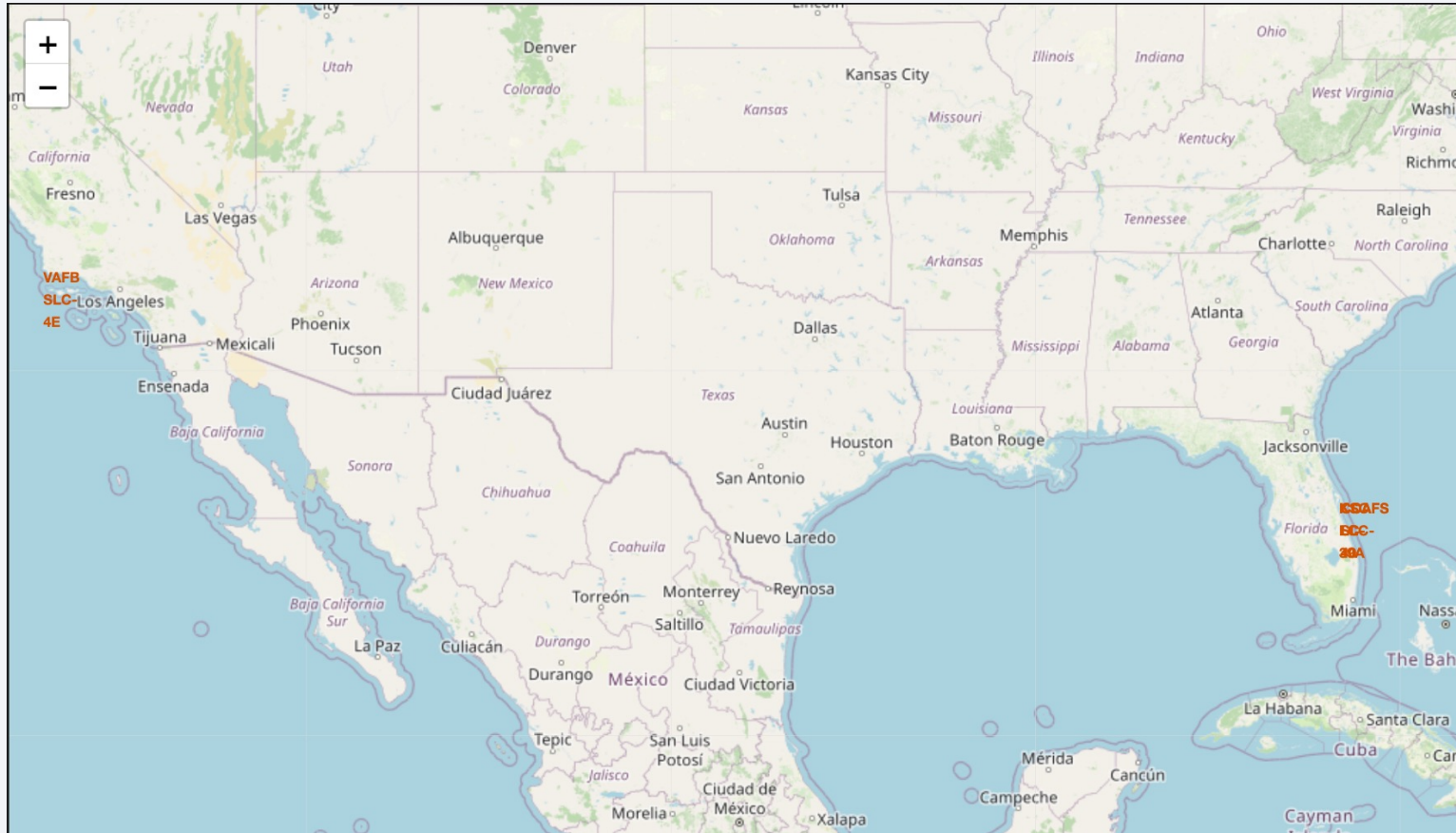
- Ranked the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order
- Used the "group by" to extract distinct "Landing_Outcome" and used "order by" function to make them descending order

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue background on the left and a satellite photograph of Earth on the right. The Earth's surface is dark, with numerous bright yellow and orange lights representing cities and urban areas. The horizon of the Earth is visible as a thin, curved line separating the dark surface from the deep blue of space.

Section 3

Launch Sites Proximities Analysis

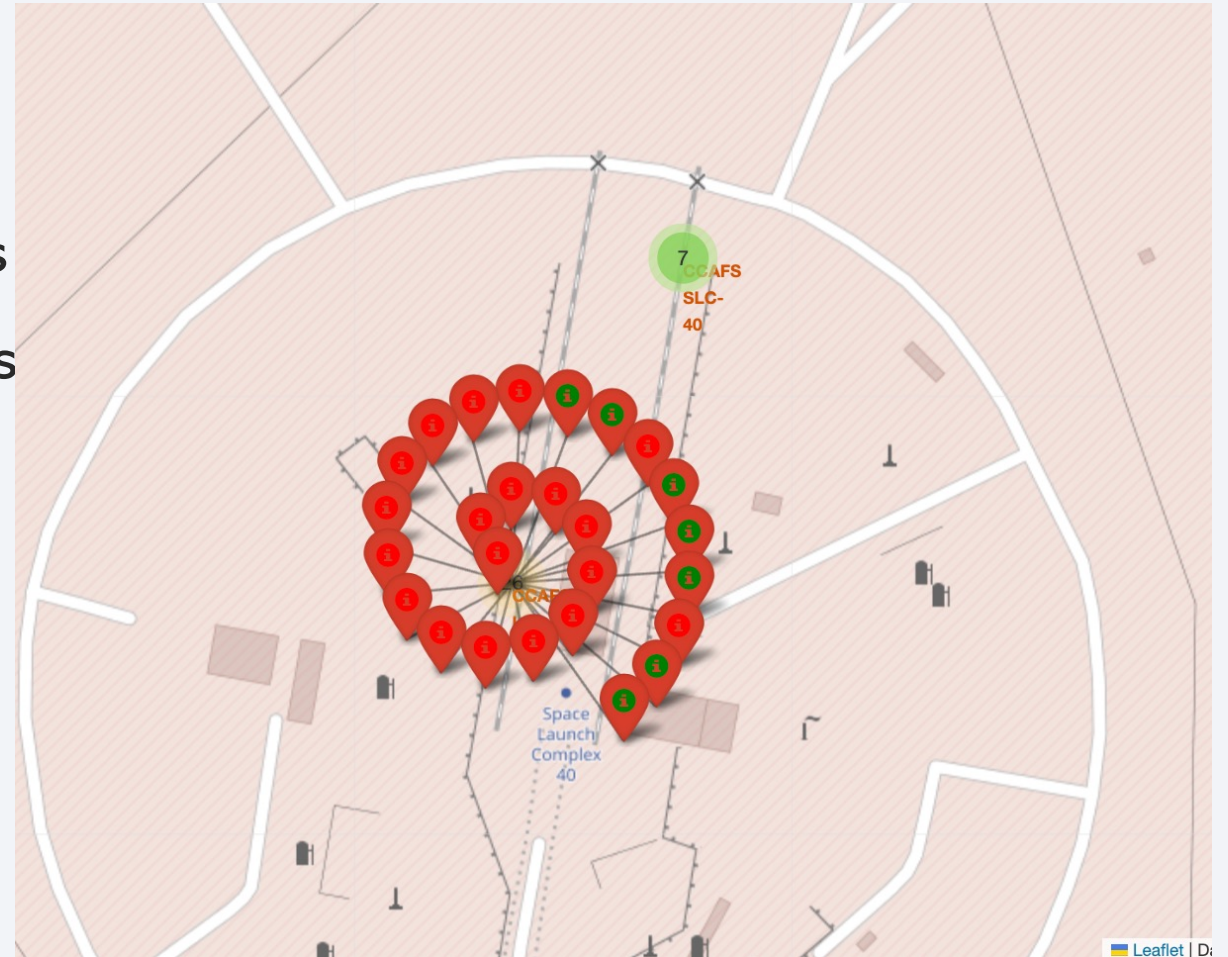
<All launch sites location markers on global map>



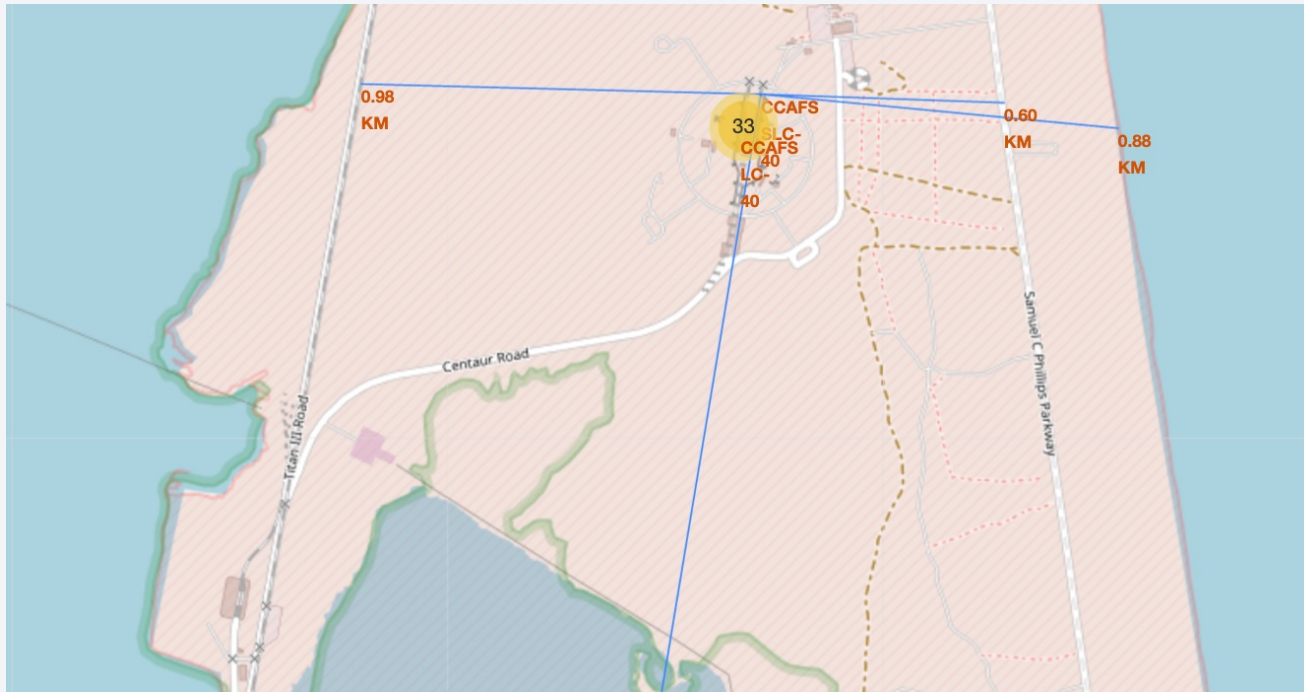
- All launch sites in proximity to the Equator line
- All launch sites in very close proximity to the coast

<Color-labeled launch outcomes>

- The color labeled green a red
it is easier to identify the success
and failure from each launch sites
- Also, the number of launches
for each launch sites



<Distance calculated to railway, highway, coastline>



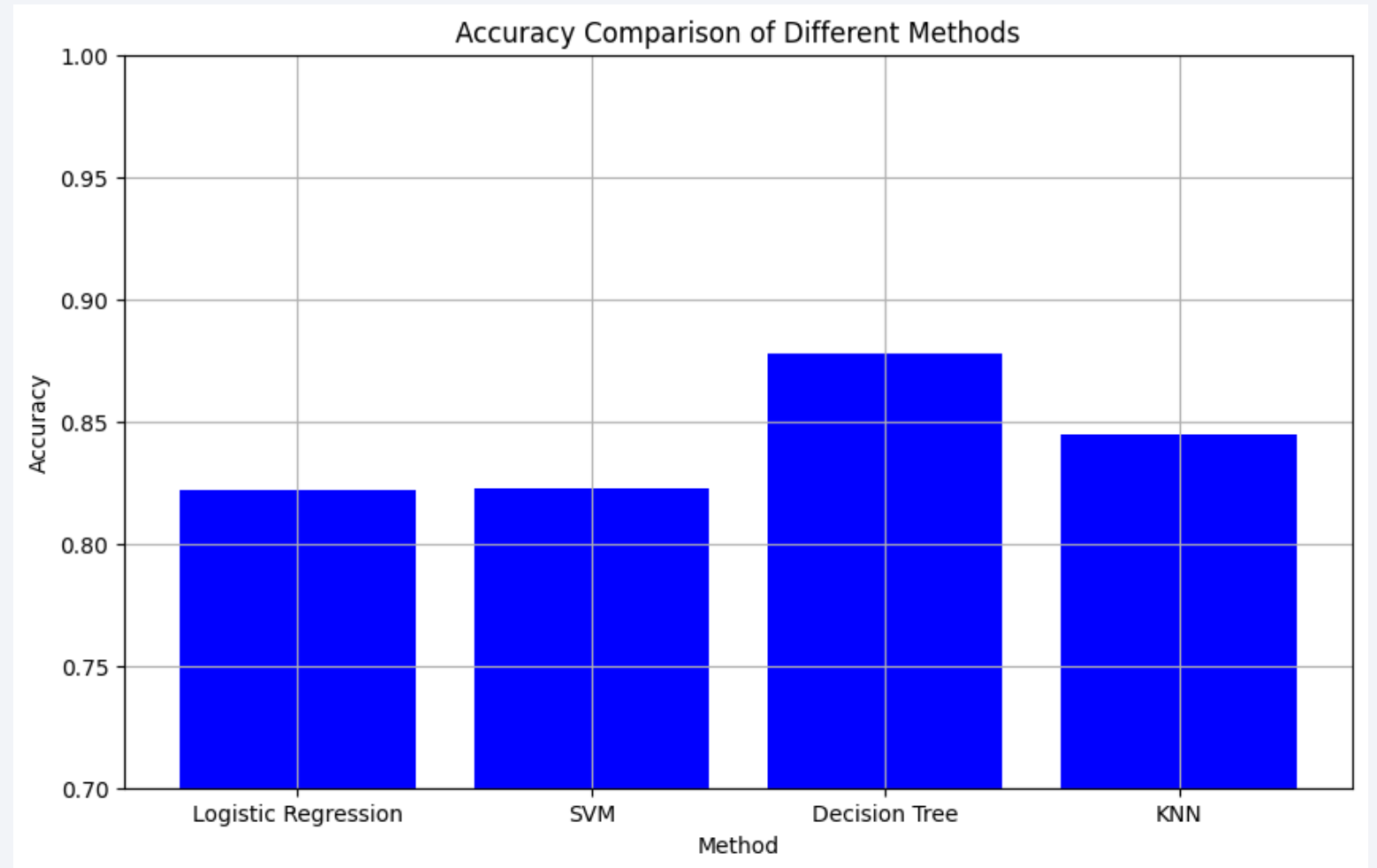
- The launch site is very close to railway, highway, and coastline
- Also, the line toward the south side is toward the closest city. It makes sure that the launch site keeps certain distance between the city

Section 5

Predictive Analysis (Classification)

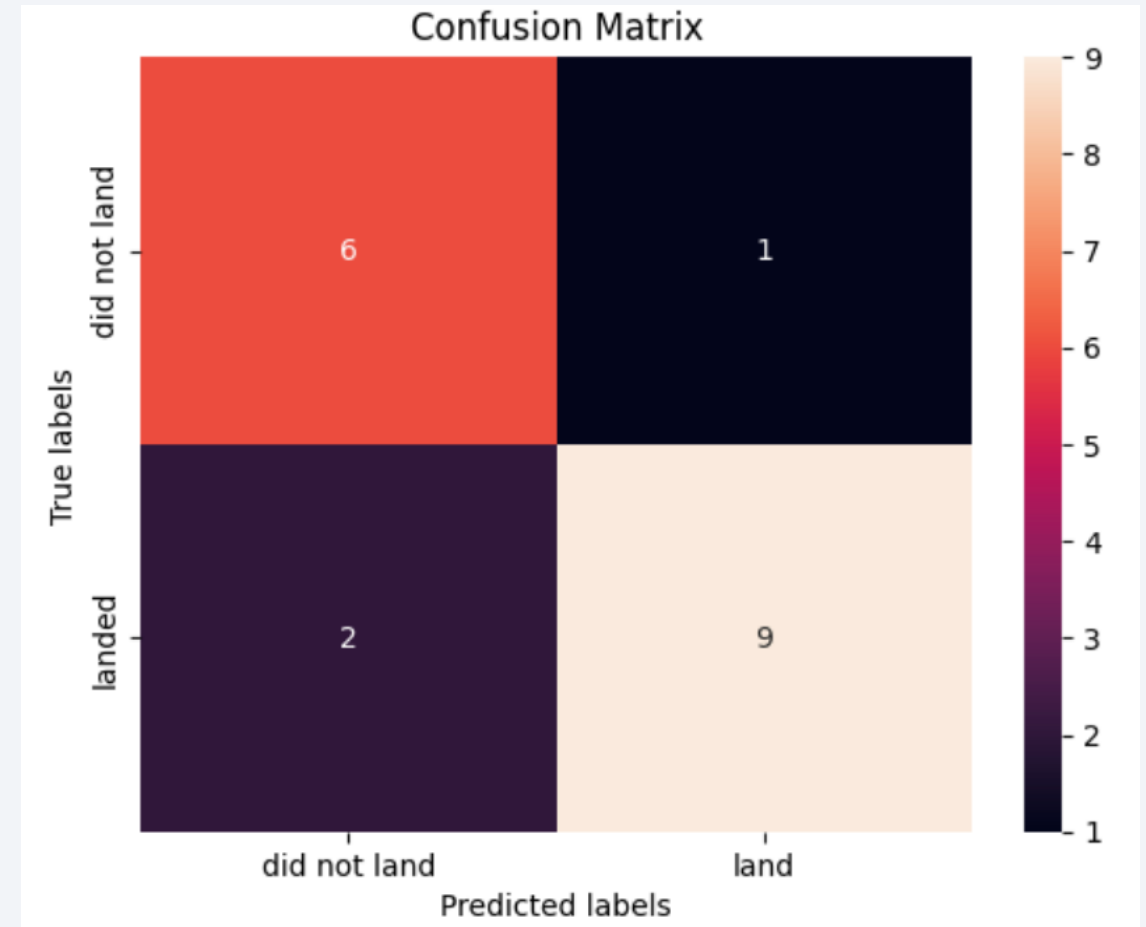
Classification Accuracy

- The best model among logistic regression, SVM, Decision Tree, KNN is the Decision Tree method.
- The decision tree method scores about 0.88



Confusion Matrix – Decision Tree

- Confusion matrix of the decision tree model
- This confusion matrix shows how accurate the true value is.
- True labels and Predicted labels with both did not land and land has much higher count than mismatch



Conclusions

- From dataset related to spaceX, shows that the success rate of the payload is related to flight number which can be considered as a year or time.
- Other than flight number, orbits, launch sites or payload mass do not seem to correlate to the success of the launches
- Among logistic regression, SVM, decision tree, and KNN, by comparing the best accuracy for prediction, the decision tree shows the best score.
- The above claim is supported by the confusion matrix of the decision tree.

Thank you!

